

Ping-Chun Lin

Seattle, WA 98115 📍 Mobile: 2063210856 ✉️ pcl1225@gmail.com 🔗 [linkedin.com/in/pingchunlin](https://www.linkedin.com/in/pingchunlin)

SUMMARY

Applied Scientist specializing in large-scale machine learning systems for real-world applications. Experience building deep learning, generative models, and production ML pipelines in PyTorch across distributed environments. Strong focus on model evaluation, uncertainty calibration, and translating research prototypes into reliable, scalable systems.

PROFESSIONAL EXPERIENCE

Fujitsu Research of America, Converging Technologies Lab Jan 2025 - Current
Research Scientist Intern Santa Clara, CA

- Designed and trained GPU-accelerated generative and foundation models (diffusion, vision transformers) for multimodal learning in PyTorch, improving performance by **10%** via systematic ablation and hyperparameter tuning.
- Implemented calibrated confidence scores to support risk-aware decision thresholds in downstream workflows.
- Collaborated cross-functionally with research and engineering teams to integrate prototype models into scalable, automated production workflows.
- Built end-to-end training and evaluation pipelines on distributed infrastructure (Slurm, CUDA), enabling reproducible experimentation across large datasets; work resulted in a **USPTO patent submission** and top-tier conference publication.

Conservation International, Betty and Gordon Moore Center for Science Jun 2024 - Jan 2025
Applied Scientist Intern Seattle, WA

- Architected and optimized a large-scale ML data processing pipeline (Python, Docker, AWS S3/EC2), reducing end-to-end runtime by **97%** through parallelization and workflow redesign.
- Designed automated feature extraction and model evaluation workflows supporting analysis across 500+ regions.

Global Innovation Fund, UW School of Environmental and Forest Sciences Jun 2023 - Sep 2023
Machine Learning Engineer Seattle, WA

- Designed and built a computer vision pipeline for object detection, segmentation, and change detection on satellite imagery using CNNs and foundation models, reducing end-to-end production time by **90%** compared to prior workflows.
- Built a reusable GeoAI software package spanning backend execution and a frontend UI, enabling transparent inspection and validation of model-generated results.

UW Applied Physics Lab, MicroSWIFT Team Sep 2022 – Dec 2022
Software Developer Seattle, WA

- Built an edge-computing data validation pipeline on Raspberry Pi devices and contributed to a production-grade Python package (PyPI) for automated wave telemetry processing.

University of Washington, Earth and Space Sciences & eScience Institute Sep 2019 – Current
Research Scientist Seattle, WA

- Designed experimental frameworks to benchmark model performance under distribution shift, including cross-validation and stress testing across heterogeneous datasets.
- Conducted error analysis and ablation studies to identify failure modes and improve model robustness.
- Authored three first-author peer-reviewed publications and secured more than **\$100K** in competitive research funding through proposal applications.

EDUCATION

University of Washington
Ph.D. in Data Science & Earth and Space Sciences 2026
M.S. in Applied Mathematics 2025
B.S. in Geophysics 2019

SKILLS

- Programming:** Python (PyTorch, TensorFlow, scikit-learn), SQL, R, MATLAB
- Machine Learning:** Deep Learning, Transformers, Generative Models, Vision-Language Models, Computer Vision, Multi-modal Learning, Model Calibration, Uncertainty Quantification, Large-Scale Training
- ML Systems & Infrastructure:** HPC (Slurm), CUDA, Distributed Data Parallel (DDP), Multi-GPU Training, Mixed Precision (AMP), Docker, AWS (S3, EC2), ETL Pipelines, Git, Unix/Linux, CI/CD
- Data Science & Visualization:** Statistical Modeling, Feature Engineering, Model Evaluation, Data Visualization, EDA, SQL Analytics (Postgres)
- Geospatial Analysis:** GDAL, Rasterio, QGIS, GeoPandas, xarray, Shapely, Google Earth Engine